SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Technical Presentation

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	**Assessment:**	Assessment Tool 3 – Technical Presentation
Weightage:	35% of CIA	**Total Marks:**	100
CO1 Statement:	*Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.*		
Student Name:	M. DHANUSH KUMAR	**Reg. No.:**	192525318
Section / Batch:		**Date:**	15/07/2026

Code of Conduct

I M.DHANUSH KUMAR (192525318)____________________ *(Name / Reg No)* *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ____________________

Instructions

- This assessment contains technical presentation. Total: 100 Marks.
- When a student delivers a technical presentation on a data structures topic, evaluation should be based on the following requirements: Concept explanation, Real-world application and Clarity of presentation.
- Demonstrate clear understanding, not just reading slides
- Use of tools: PowerPoint / Google Slides.
- Duration: 7–10 minutes presentation + 3 minutes Q&A.

Section / Topic	Focus	Q Nos	Marks
Data Structure	Real-Time Applications	1	100
		GRAND TOTAL:	**100 Marks**

Annexure A – Technical Presentation

A web browser supports navigating backward through previously visited pages. Recommend a suitable ADT and prepare a technical presentation explaining how it improves performance efficiency. (100 Marks)

(OR)

A music streaming application maintains a playlist where songs can be added, removed, or reordered dynamically. Recommend a suitable data structure and justify based on flexibility and efficiency. Prepare a technical Presentation. (100 Marks)

(OR)

A metro rail network maps stations and routes to determine reachable stations from a source station. Suggest a suitable data structure for representation and justify your answer. (100 Mark)

(OR)

A calculator application evaluates arithmetic expressions entered by users. Recommend a suitable data structure and justify based on expression processing efficiency. (100 Marks)

(OR)

A cafeteria token system serves customers strictly in the order of arrival. Identify the suitable ADT and justify why it ensures fairness. (100 Marks)

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	20	Demonstrates deep and accurate understanding of data structure with insights and connections	Shows clear understanding with minor gaps	Basic understanding; some misconceptions present	Limited or incorrect understanding
Content Organization	30	Logical, well-structured, smooth flow; strong introduction and conclusion	Mostly organized with clear flow	Some organization but lacks clarity or transitions	Disorganized, difficult to follow
Technical Depth	20	Includes algorithms, complexity analysis, and advanced insights	Includes algorithm and basic complexity analysis	Limited technical details; missing depth	Minimal or no technical content
PowerPoint Slides	20	Excellent design, clear visuals, enhances understanding	Good slides with minor issues	Basic slides; somewhat cluttered or unclear	Poor slides or no visual support
Communication Skills	10	Clear, confident, engaging delivery; excellent articulation	Clear delivery with minor hesitation	Some hesitation; unclear in parts	Poor communication; difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: ____________	Signature: ____________	Signature: ____________
Date: ________________	Date: ________________	Date: ________________

CSA0305– DATA STRUCTURE

A metro rail network maps stations and routes to determine reachable stations from a source station.

PRESENTED BY:

M,Dhanush kumar(192525318)

GUIDED BY:

Dr.Uma Priyadharshini

ABSTRACT

- A metro rail network can be efficiently represented using a graph data structure.
- In the graph, stations are vertices (nodes) and routes are edges connecting them.
- The graph helps identify all reachable stations from a given source station.
- Breadth-First Search (BFS) and Depth-First Search (DFS) are used to traverse the network.
- This approach improves route planning, connectivity analysis, and network management.

INTRODUCTION

- A metro rail network consists of multiple stations connected by railway routes.
- Each route between two stations is represented as an **edge**.
- A graph data structure efficiently represents the connectivity of the network.
- It helps determine all reachable stations from a given source station.
- This representation is widely used in real-world transportation and navigation systems.

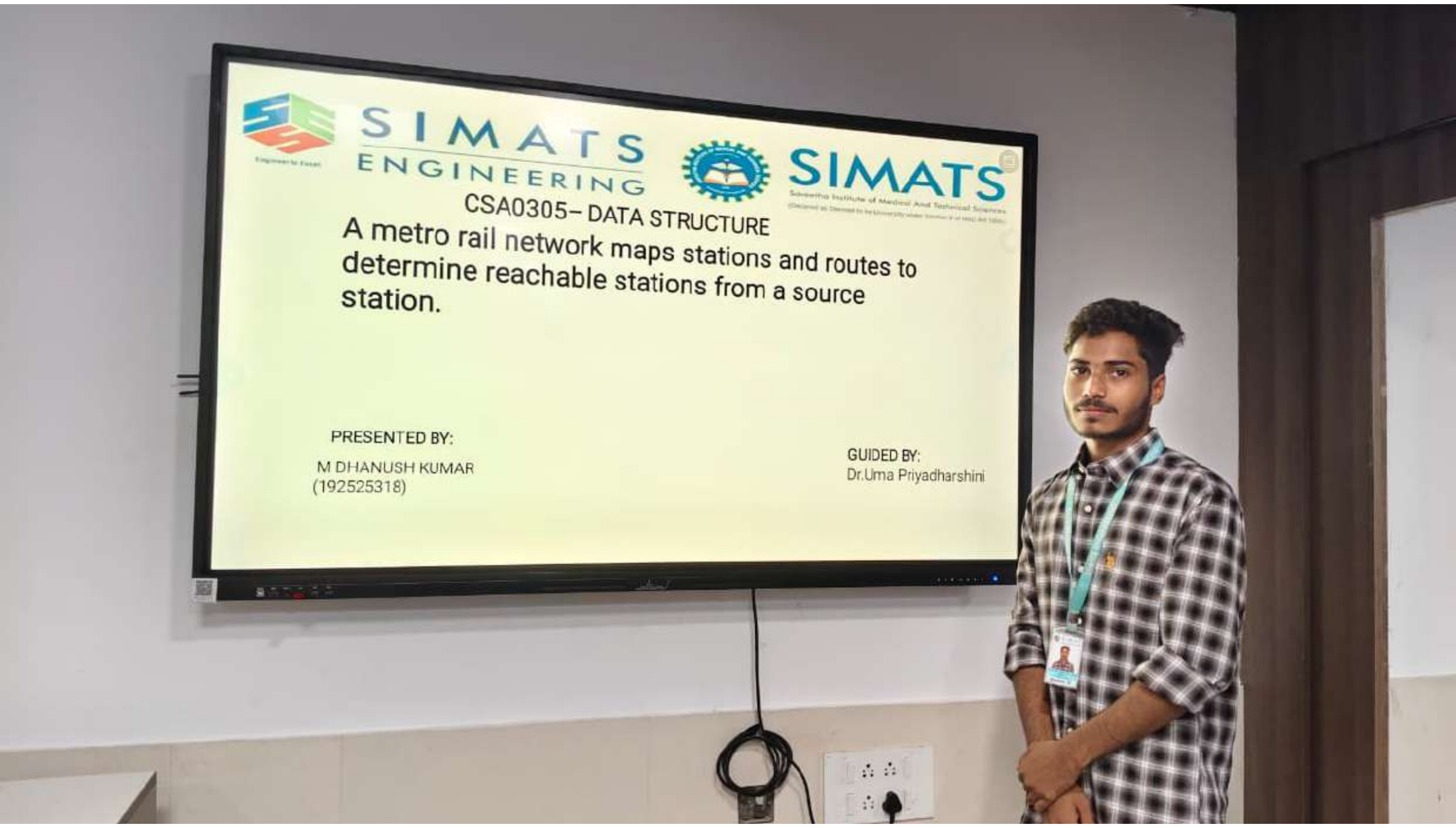
FUTURE ENHANCEMENT

- Add shortest path algorithms to find the fastest route between stations.
- Include real-time train schedules and delay information.
- Support multiple metro lines and interchanges.
- Integrate GPS and live tracking for better navigation.
- Develop a mobile application for route planning and ticket booking.

CONCLUSION

- A graph data structure is the most suitable way to represent a metro rail network.
- It efficiently models stations and routes as nodes and edges.
- BFS and DFS help identify all reachable stations from a source station.
- The graph approach improves route planning, connectivity analysis, and network management.
- It is simple, efficient, and widely used in modern transportation systems.

Engineer to Excel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